The LHS 1140 b catalog: what is certified and what is not

The physics of this summary. Section 5 (the molecular $K_{zz} = 10^9$ family) was rewritten on 2026-09-26 from states certified under the atomic data as corrected on 2026-09-24 to 2026-09-26 (`md/Update_EXHALE_stage3.md` sections 17 to 27). Every other state, equivalent width, crossing and certificate below is of the physics before those corrections: none of those certificates holds under the corrected physics, and their equivalent widths are not current (the molecular family's crossing moved from He/H = 1.46 to 1.70 on the catalog grid). **The code moved again on 2026-09-26 to 2026-09-28** (sections 29.1 to 29.12 of the same log: an OpenMP race in the metal sweep, the interdiffusion enthalpy flux, the $He^+ + H$ charge exchange near threshold, the `_adv` post-process made second order, the lower face placed at the reservoir level): every number of this summary, section 5 included, predates those changes. **On 2026-09-29 the code moved once more** (sections 33 to 37): the sensible enthalpy the transported molecular carriers carry is in the energy equation (default on), and under `Well balanced: True` the base boundary shares the reconstruction's discrete hydrostatic equilibrium (boundary model `. . . _discrete_equilibrium_v5`), which removes a base mass flux of up to 10^4 times a weak wind. Two continuation states certified under the previous boundary re-certified under it in 2 and 4 passes (MEASURED). By the user's decision of 2026-09-26 certified rows are not re-solved after each change; the models that did not solve come first (section 3). **From 2026-09-30 to 2026-10-05 the equation set itself changed** (sections 39 to 102 of the same log): heat conduction with the conductivity of the mixture (`Conduction: True`), the ionization stages carried as transported species (`Ionization transport: True`) with their enthalpy, He 2^3S carried as a fourth helium state (`He 2^3S transport: True`), the advected product stepping each element on the velocity of its own nuclei, and a Photochem column coupled to the wind in energy as well as in the elemental fluxes. No catalog row was re-solved under these changes; the inventory, re-run read-only on 2026-10-05, is unchanged. Heat conduction alone lowers the mass-loss rate of the three fiducial-XUV catalog cases re-solved with it by factors 0.38 to 0.69 (log sections 48, 51 and 52, taken from the log), and their He 10830 EWs move with it (sections 49 to 61), so every EW and crossing of sections 3 to 5 is of a model without heat conduction. The models solved with the full equation set stand outside the catalog, in section 6..

Every model of the catalog is LHS 1140 b on 500 cells, solved by the partitioned stationary route with `Well balanced: True`, from the GJ 1132 spectrum placed at this planet's orbit. The binary of record of the rows solved up to 2026-09-22 is `EXHALE_75d55d9d.x`, md5 `75d55d9d4fd0e748cd01d6e35713e34c`; the rows added on 2026-09-23 and 2026-09-24 were solved and certified under `EXHALE_69233669.x` (the He/H = 9.7 rungs below 0.07 of the fiducial XUV) and `EXHALE_46d5b9f4.x` (the molecular $K_{zz} = 10^9$ rungs), as the table notes say. Five molecular $K_{zz} = 10^9$ rows were certified after the atomic-data corrections began: He/H = 1.45 on 2026-09-25 under `EXHALE_b059a4be.x` (the second of five rounds of corrections), and 1.52, 1.6, 1.8 and 2.13 on 2026-09-26 under the binary of record, `EXHALE_7db5afe9.x`, md5 `7db5afe9b1d80ce757fcc07d1b499330` (section 5). Three molecular rows were published on 2026-09-29 from states the continuation of `md/Update_EXHALE_stage3.md` sections 29.13, 34 and 37 certified: the photochemical ph otochem_kzzprofile/HeH9 and photochem_kzzprofile/HeH2.09 under `EXHALE_6dd54955.x` (base boundary model v4, with the enthalpy flux of the transported carriers) and scalar_wellmixed/HeH0.55 under `EXHALE_3c22c63e.x` (the discrete-equilibrium boundary model v5), as the table notes of section 4 say. The counts below are MEASURED by

`models/status.py inventory`, re-run read-only on 2026-09-29 and again on 2026-10-05 (the same counts), which names `EXHALE_6dd54955.x`, md5 `6dd549556910351a7f857708b4853a69`, as the binary of record: **210 state indexes, 155 certified** (158 and 103 earlier on 2026-09-29, before the continuation rungs were published; 157 and 100 on 2026-09-24 and 2026-09-26; 143 and 92 on 2026-09-22). These counts are of the recorded certificates: the molecular rows published on 2026-09-29 were measured under the binaries their table notes name (section 4), all other rows under binaries before the atomic-data corrections. A case counts as certified when its index names a `latest_certified` generation that no refused re-evaluation has marked stale. Section 3 and section 4 list the 156 models of the ladders and of `diffusion_check`; the remaining 54 indexes are 46 flux-closure iterates, two archival copies, one study directory and five run directories the low-XUV rungs left. Section 5, rewritten on 2026-09-26 under the corrected atomic physics: the molecular $K_{zz} = 10^9$ family straddles the observed equivalent width with two certified rungs (He/H = 1.6 and 1.8), the crossing moves with the grid to about 1.84 in the continuum limit, and the added Gaussian those solutions need to reproduce the width of the measured line is recorded.

Where each column comes from. du and $\|R\|$ are both read from the LAST `(JFNK) done` line of the case's `run.log`, so they are one pair printed by one solve and not two numbers from two places; where a case was last re-measured by an evaluation that took no step, that line is the solve preceding it, and where a case was never advanced there is none and the columns read `--`. The refusing rows of section 3 are read from `certification.txt` of the generation the case's index names `latest_complete`, and EVERY refusing entry of that certificate is listed, not only the first. $\log_{10}\dot{M}$ and the equivalent width are from `models/status.py`, which reads them from the case directory. The equivalent width is the red-pair integral over the vacuum window 10832.60 to 10834.20 Å taken on the observation's own wavelength samples: the instrument-convolved model is interpolated onto the 44 samples of the measurement inside the window and integrated with their trapezoid weights, the same operator that gives the observed 1.108 % Å (`LHS1140b/he10830_equivalent_width.py`, adopted 2026-09-23). Every EW of the tables of sections 3 and 4 is that window EW of the instrument-convolved model with no added velocity kernel ($f = 0$, the `EW=` field of an `EW.txt`); the width-matched EW, the convention adopted on 2026-10-04 for every comparison with an observed line (the model's excess absorption convolved with a Gaussian of FWHM f in velocity, f chosen so that the red-pair FWHM is the measured 0.841 Å), is quoted with its f in sections 5 and 6.. Integrated on its own finer grid instead, a model differs by at most 1.5×10^{-4} % Å over the catalog, which moves 15 of the 91 certified rows below in the fourth decimal (MEASURED 2026-09-24). The XUV column is the factor the stellar spectrum is scaled by; K_{zz} is the eddy coefficient of the base, *profile* where a photochemical column supplies $K_{zz}(p)$, and *none* where the base is well mixed. The *wind chemistry* column of every table below says what is solved ABOVE the base, atomic or molecular in the sense of section 1; what supplies the base itself is in the case name (`scalar` or `photochem`) and in the K_{zz} column.

1. Two axes: where the lower atmosphere comes from, and what the wind solves

A case name carries both axes, and they are independent. The first says what supplies the state at the base level, the second whether the wind above it carries a molecular network at all.

The lower boundary.

- `scalar`: the base is a few prescribed numbers at the level, read from `base.inp`: the pressure (1 microbar), the temperature (226 K), the base radius, the bound hydrogen fraction $q_{\mathrm{H2,base}}$ and the element reservoirs. Nothing below the level is modeled.
- `scalarCNO`: the same scalar base with the C, N and O reservoirs of the photochemical column added at the matching level.
- `photochem`: the lower atmosphere is a Photochem column handed over as a PROFILE, the key `Lower atmosphere profile:` naming a file `lower_atmosphere_profile.dat`, which carries $T(p)$, $K_{zz}(p)$, the bound hydrogen fraction and the element reservoirs at the matching level p_{match} = 1 microbar. For the atomic groups the reservoir He/H is then solved by the elemental-flux closure (`src/utils/element_flux_closure.py`), which iterates the column and the wind against each other: the iterates are the `k<NN>` directories, and the composition is an output of that loop rather than an input.

The wind's chemistry.

- `atomic`: hydrogen and helium with their ionization stages and, where the group says so, trace metals. No molecular network is solved above the base, whatever the base carries.
- `molecular`: EXHALE's own network, `Molecular chemistry: True` with `Molecular carrier transport: True`, solving H_2, H_2^+, H_3^+ and HeH^+ in the wind with the H_2 carrier transported. The molecular base it starts from is the scalar handoff or the profile, by the other axis.

So `atomic_photochem_...` is a Photochem column under an atomic wind, and `molecular_scalar_...` is EXHALE's own molecular chemistry above a scalar base. The two are not alternatives for the same layer: the Photochem column is the atmosphere BELOW the matching level, and EXHALE's network is the chemistry ABOVE it. A `molecular_photochem` case uses both, the column below and the network above.

	wind solved atomic	wind solved with EXHALE's molecular network
scalar base (`base.inp`)	`atomic_scalar_*`	`molecular_scalar_*`
Photochem column (profile)	`atomic_photochem_*`	`molecular_photochem_*`

The mixing suffix is a third, smaller axis: `wellmixed` is no element diffusion, `kzz1e9` is binary H/He diffusion at that eddy coefficient, and `kzzprofile` takes $K_{zz}(p)$ from the column.

2. By group

group	wind chemistry	He/H	XUV	K_{zz}	diff.	*du* reached	cases	cert.
diffusion_check	atomic	0.55	1	0.0 to 1.0e9	on	–	4	0
photochem_x0.01_kzzprofile	atomic	9.7	0.01	profile	on	9.4E-03	1	0
photochem_x0.10_kzzprofile	atomic	9.7	0.10	profile	on	5.0E-10	1	1
photochem_x0.15_kzzprofile	atomic	9.7	0.15	profile	on	4.7E-12	1	1
photochem_x0.20_kzzprofile	atomic	9.7	0.20	profile	on	2.0E-12	1	1

group	wind chemistry	He/H	XUV	K_{zz}	diff.	*du* reached	cases	cert.
photochem_x0.25_kzzprofile	atomic	9.7	0.25	profile	on	1.9E-12	1	1
photochem_x0.30_kzzprofile	atomic	9.7	0.30	profile	on	1.3E-10	1	1
photochem_x0.33_kzzprofile	atomic	9.7	0.33	profile	on	1.6E-10	1	1
scalarCNO_kzz1e9	atomic	2.13	1	1.0e9	on	1.7E-11	1	1
scalar_gj699_wellmixed	atomic	0.042 to 1000	1	none	off	1.8E-13 to 1.6E-11	6	6
scalar_kzz0	atomic	0.55 to 3.9	1	0.0	on	2.4E-11 to 1.2E-10	6	6
scalar_kzz1e10	atomic	0.55 to 1.29	1	1.0e10	on	2.0E-11 to 2.7E-11	4	4
scalar_kzz1e11	atomic	0.55 to 0.93	1	1.0e11	on	1.8E-11 to 1.9E-10	4	4
scalar_kzz1e5	atomic	2.6 to 3.93	1	1.0e5	on	2.5E-11 to 4.8E-11	5	5
scalar_kzz1e6	atomic	0.55 to 3.74	1	1.0e6	on	1.9E-11 to 6.8E-11	6	6
scalar_kzz1e7	atomic	0.55 to 3.18	1	1.0e7	on	1.6E-12 to 1.3E-10	4	4
scalar_kzz1e8	atomic	0.55 to 2.41	1	1.0e8	on	5.5E-11 to 7.9E-11	4	4
scalar_kzz1e9	atomic	0.55 to 9.7	1	1.0e9	on	9.3E-13 to 6.1E-11	7	7
scalar_wellmixed	atomic	0.083 to 1000	1	none	off	1.4E-12 to 1.2E-10	9	9
scalar_x0.015_kzz1e9	atomic	2.13	0.015	1.0e9	on	1.8E-10	1	1
scalar_x0.01_kzz1e9	atomic	2.13, 9.7	0.01	1.0e9	on	4.4E-12 to 2.2E-10	2	1
scalar_x0.02_kzz1e9	atomic	2.13	0.02	1.0e9	on	8.3E-11	1	1
scalar_x0.03_kzz1e9	atomic	2.13	0.03	1.0e9	on	4.4E-11	1	1
scalar_x0.05_kzz1e9	atomic	2.13	0.05	1.0e9	on	6.4E-11	1	1
scalar_x0.06_kzz1e9	atomic	9.7	0.06	1.0e9	on	2.8E-12	1	0
scalar_x0.0624_kzz1e9	atomic	9.7	0.0624	1.0e9	on	3.1E-12	1	0
scalar_x0.065_kzz1e9	atomic	9.7	0.065	1.0e9	on	2.6E-12	1	1
scalar_x0.0662_kzz1e9	atomic	9.7	0.0662	1.0e9	on	2.5E-12	1	1
scalar_x0.0675_kzz1e9	atomic	9.7	0.0675	1.0e9	on	3.3E-12	1	1
scalar_x0.07_kzz1e9	atomic	2.13, 9.7	0.07	1.0e9	on	7.0E-12 to 5.1E-11	2	2
scalar_x0.10_kzz1e9	atomic	2.13, 9.7	0.10	1.0e9	on	4.4E-12 to 1.5E-11	2	2
scalar_x0.15_kzz1e9	atomic	2.13, 9.7	0.15	1.0e9	on	8.2E-12 to 9.1E-12	2	2
scalar_x0.20_kzz1e9	atomic	2.13, 9.7	0.20	1.0e9	on	5.5E-12 to 1.1E-11	2	2
scalar_x0.25_kzz1e9	atomic	2.13, 9.7	0.25	1.0e9	on	5.2E-12 to 1.7E-11	2	2
scalar_x0.30_kzz1e9	atomic	2.13, 9.7	0.30	1.0e9	on	3.2E-12 to 5.6E-12	2	2
scalar_x0.33_kzz1e9	atomic	2.13, 9.7	0.33	1.0e9	on	2.8E-12 to 3.4E-12	2	2
photochem_kzzprofile	molecular	2.09, 9	1	profile	on	3.6E-12 to 4.9E-12	2	2
scalar_kzz1e9	molecular	0.083 to 9.7	1	1.0e9	on	8.2E-13 to 1.2E-10	40	39
scalar_wellmixed	molecular	0.083 to 2.13	1	none	off	7.8E-13 to 2.5E-10	22	21
total							156	146

Besides these, the catalog holds 46 flux-closure iterates k<NN>, two archival copies, one study directory and five run directories of the low-XUV rungs (each run keeps the case files it ended with, and its index counts in the inventory). Nine of the iterates carry a certificate; the other 37 were written by runs that asked for no time integration, so they are states that were post-processed and never advanced, and they are not failures of a solve.
One molecular case carries no index at all and is not counted in the table: molecular_scalar_gj1132_wellmixed/HeH2 .13 was attempted and published no state (section 3). molecular_photochem_gj1132_kzzprofile/HeH2.09, which was in the same position, has published a certified state since 2026-09-29 (section 4).

3. The models that are not certified

Ten of the 156 indexed models carry no certificate. Four of them, the `diffusion_check` rows, are not models of their own: each is the post-processing pass of a certified case repeated with `EXHALE_DIFFUSION_CHECK=1` (`Do only PP: True`, CFL 10^{-12}; `models/diffusion_check/README.md`) to write the element face-flux table, and it makes no stationary claim (`cert_reason=no_stationary_claim`). The four cases they repeat, the `HeH0.55` rungs of `atomic_scalar_gj1132_kzz0`, `_kzz1e8`, `_kzz1e9` and `_kzz1e10`, are certified (`certified_in_wind`, read 2026-09-29), and the rows the certificate refuses on below are those of the post-processed state, not of a solve. The other six were solved, or attempted, or written without being advanced at all, and the certificate of the generation its index names refuses them. The last column lists every entry of that certificate that stands above its tolerance, with the measure and the cell, in the certificate's own order.

case	wind chemistry	He/H	XUV	K_{zz}	diff.	*du*	$\|\|R\|\|$	the rows the certificate refuses on
diffusion_check/scalar_kzz0_HeH0.55	atomic	0.55	1	0.0	on	–	–	mass 1.541 at cell 1; momentum 7.593e-3 at cell 500; energy 1.583 at cell 5; elemental He/H 2.546e-4 at cell 500
diffusion_check/scalar_kzz1e10_HeH0.55	atomic	0.55	1	1.0e10	on	–	–	mass 1.541 at cell 1; momentum 2.466e-3 at cell 500; energy 1.907 at cell 5; elemental He/H 9.644e-5 at cell 500
diffusion_check/scalar_kzz1e8_HeH0.55	atomic	0.55	1	1.0e8	on	–	–	mass 1.542 at cell 1; momentum 5.150e-3 at cell 500; energy 1.952 at cell 5; elemental He/H 1.790e-4 at cell 500
diffusion_check/scalar_kzz1e9_HeH0.55	atomic	0.55	1	1.0e9	on	–	–	mass 1.542 at cell 1; momentum 3.719e-3 at cell 500; energy 1.917 at cell 5; elemental He/H 1.354e-4 at cell 500

case	wind chemistry	He/H	XUV	K_{zz}	diff.	du	$\|R\|$	the rows the certificate refuses on
photochem_x0.01_kzzprofile/HeH9.7	atomic	9.7	0.01	profile	on	9.356E-03	1.163E+00	mass 5.880e-1 at cell 11; momentum 7.774e-7 at cell 495; energy 1.949 at cell 7; elemental He/H 3.538e-2 at cell 389
scalar_x0.01_kzz1e9/HeH9.7	atomic	9.7	0.01	1.0e9	on	4.424E-12	9.978E-01	mass 9.806e-1 at cell 2; momentum 8.724e-6 at cell 1; energy 1.071 at cell 1
scalar_x0.06_kzz1e9/HeH9.7	atomic	9.7	0.06	1.0e9	on	2.785E-12	5.589E-07	elemental He/H 2.508e-2 at cell 333
scalar_x0.0624_kzz1e9/HeH9.7	atomic	9.7	0.0624	1.0e9	on	3.086E-12	8.091E-08	elemental He/H 1.435e-2 at cell 333
scalar_kzz1e9/HeH0.083	molecular	0.083	1	1.0e9	on	4.879E-12	1.369E-08	H_2 carrier 8.150e-4 at cell 499; elemental He/H 2.503e-4 at cell 280
scalar_wellmixed/HeH0.083	molecular	0.083	1	none	off	6.592E-12	2.701E-08	H_2 carrier 7.371e-2 at cell 500

Read by mechanism, the ten fall into three kinds.

- **The four atomic He/H = 9.7 models at 0.01 to 0.0624 of the fiducial XUV** (the hydrodynamic Newton made no progress at 0.01; the elemental He/H row refused 0.06 and 0.0624 at cell 333). Their catalog states lie on the grid before the lower face was placed at the level, so they load only as seeds mapped onto the present grid, and there the stationary solve fails on every pass (`info = 2`). The mapped seeds are winds (mass-flux spread 0.3 per cent), but their lower layer is off the grid's discrete hydrostatic equilibrium by about 10^{-3} of the pressure, which the Roe pressure-jump term turns into a numerical mass flux of up to 10^4 times the wind; projected onto that equilibrium (with the base boundary sharing it), the mass and momentum rows fall from 1.9 and 5×10^{-3} to 3.7×10^{-3} and 8×10^{-7} on the 0.06 seed, and the energy row, about 1, is what remains: the heat of the base cell and the interdiffusion enthalpy flux of a composition that is not in its diffusive steady state near the base (`md/atomic_heh97_reduced_xuv_20260929.md`, its review, and `md/Update_EXHALE_stage3.md` sections 35 to 38). From the projected seeds, with the element interdiffusion enthalpy flux off, 0.06 and 0.0624 then certified, and continuation chains switched on heat conduction by its factor (sections 38 to 47; taken from the log): the chains were stopped by hand on 2026-09-30 at 23:57 with the last certified rungs at conduction factor 0.668 (0.06) and 0.751 (0.0624) and the photochemical XUV descent at 0.0833 of the fiducial XUV (`md/session_handoff_20260930.md` section 8; the later rungs in `.P1/atomic97/` end `NOT CERTIFIED` in their `run.log`, READ 2026-10-05). None of these states is published to the catalog. With conduction the photochemical case at 0.10 of the fiducial XUV becomes a conductive thermosphere below an exobase at 1.5 R_p, outside the validity of the hydrodynamic solution (log section 45).
- **The composition half of the solve refuses** the two molecular models: the H_2 carrier balance in both, and in sc alar_kzz1e9/HeH0.083 also the elemental He/H row. The wind itself is converged in both. The next subsection says what was measured. Two more molecular models stood in this table until 2026-09-29, photochem_kzzp rofile/HeH9 and scalar_wellmixed/HeH0.55; they were reached by continuation and their certified states are published (section 4). The two that remain are approached to He/H = 0.19 and 0.1 (the subsection after next).
- **Nothing was solved** in the four `diffusion_check` rows: they are post-processing passes of four certified cases, made without a time integration, so they have no du and no $\|R\|$. Their certificates were measured all the same, and refuse on the hydrodynamic rows and the elemental row, which is what an unadvanced state is expected to do; they are not failures of a solve, and not models of their own.

Stale (2026-10-07). The Knudsen numbers, $\mathrm{Kn}_{\mathrm{bulk}} = 0.1$ radii and exobase radii in this passage were computed before the resonant $H^+ + H$ and $He^+ + He$ momentum-transfer coefficients entered `collisional_validity.py` (Update_EXHALE_stage3 section 110). On the 102 stored LHS 1140 b states the $\mathrm{Kn}_{\mathrm{bulk}} = 0.1$ radius moves out by 1.0–5.9 %, the exobase out by 0.4–5.7 % and the largest Kn down by 0–16 %; $\mathrm{Kn}(H_2)$ and the sonic point do not move and no verdict changes. Not re-measured for the states quoted here.

What the two numbers say. du is the fractional spread of the mass flux $\rho v r^2$ over the wind window, the quantity a marching run stops on, and $\|R\|$ is the stationary operator's residual norm. The recipe of this code, and the practice of the literature it follows, calls a wind converged at $du < 10^{-3}$. MEASURED on the models that carry no certificate, naming the wind chemistry because the same short name occurs in both families: the molecular scalar_kzz1e9/He H0.083 reaches $du = 4.9 \times 10^{-12}$ with $\|R\| = 1.4 \times 10^{-8}$, the molecular scalar_wellmixed/HeH0.083 6.6×10^{-12} and 2.7×10^{-8}, and the atomic scalar_x0.01_kzz1e9/HeH9.7 4.4×10^{-12} with $\|R\| = 1.0$, and the atomic He/H = 9.7 rungs at 0.06 and 0.0624 2.8×10^{-12} and 3.1×10^{-12} with 5.6×10^{-7} and 8.1×10^{-8}. By the flux-spread criterion these are converged winds, and two of them also sit at a residual norm of 10^{-8}.
What refuses them is this code's own row-by-row certification, which is stricter: it measures every equation of the inventory cell by cell and names each one that stands above its tolerance. Over the ten the elemental He/H row appears

in eight of them, the hydrodynamic energy, mass and momentum rows in six each and the H_2 carrier balance in the wind in two (MEASURED on the table above). A state that meets the flux-spread criterion and fails one such row is a meaningful solution by the standard other work uses, and the table says which rows keep it from this one.

Why the molecular winds do not certify

The verdict. The failure is of the method, not a finding that these winds have no stationary state. The partitioned route alternates two solves in every outer pass: the hydrodynamic Newton solve at fixed composition, then the relaxation of the composition (the element rows and the H_2 carrier) at fixed wind. For the six molecular cases below the alternation does not approach a joint fixed point. The one route that solves both halves together fails in its linear solve. Every measurement of this subsection was made under binaries before the atomic-data corrections, among them `EXHALE_3146d11b.x` (2026-09-18) and the tree's binary of 2026-09-21 (md5 `89ec6714...`). Whether the same cases behave the same way under `EXHALE_7db5afe9.x` was not measured.

The six cases. The two molecular rows of the table above, the two that stood in it until they certified by continuation (photochem_kzzprofile/HeH9 and scalar_wellmixed/HeH0.55, section 4), and the two that published no index when these measurements were made (READ from each case's `not_solved.md`, now `not_solved_<date>.md` where the case has solved since, and from `md/lhs1140b_stationary_L34c_20260918.md`); of those two, photochem_kzzprofil e/HeH2.09 has certified by continuation as well. The two with no index were started from the certified atomic wind of the same composition. Each was stopped at the six-hour ceiling:

- molecular_scalar_gj1132_wellmixed/HeH2.13: the carrier H_2 row flattened at 4.3×10^{-1} after seven outer passes.
- molecular_photochem_gj1132_kzzprofile/HeH2.09: the row banded between 0.19 and 0.44 over twelve outer passes without a trend.

What six outer passes showed, measured on 2026-09-21. Each of four cases was advanced six outer passes from its stored state (`md/lhs1140b_alternation_audit_20260921.md`). Three did not approach a fixed point, by two different mechanisms, and one did:

- **The composition update takes the wind out of solution** (wellmixed/HeH0.083 and wellmixed/HeH0.55).
  - One bounded composition update moved the mass flux by 0.985 of its own column scale.
  - From pass 2 on, the hydrodynamic solve returned `info = 2`, with its mass and energy rows seven decades above tolerance, and never recovered.
  - In wellmixed/HeH0.55 the H_2 front also walked outward: the refusing cell moved from 306 to 349 (r = 1.95 to 3.02 R_p) while the row rose by a factor 4.5.
- **A drift under a saturated movement bound** (kzz1e9/HeH0.083).
  - The carrier relaxation ended on its movement bound in every pass, at cells 418 to 426 (r = 8.0 to 8.7 R_p).
  - The gated carrier row rose from 8.2×10^{-4} to 2.9×10^{-3}, while the hydrodynamic rows stayed within tolerance.
  - Relocating the bound does not help: the bound then acts at the edge of the mask, and the refusing row rises in every such run (2026-09-17). So the bound is not the defect.
- **Approaching, and stopped by the budget** (kzz1e9/HeH0.55): the joint residual fell 3.7 decades in five passes. This case was then finished and certified on 2026-09-24 (section 4).

The route that solves both halves together fails. Two measurements show it, on 2026-09-17 and 2026-09-21 (`md/lhs1140b_coupled_trial_20260921.md`). With `Coupled carrier solve: True`, the element and H_2 carrier rows become unknowns of the Newton solve.

- On kzz1e9/HeH0.083, that solve spent 41 trust-region iterations, 6632 residual evaluations and 7172 s. It moved the merit only from 9.720×10^{-2} to 9.718×10^{-2}, stagnated (`info = 2`) and handed back the state it started from.

- On a case the alternation certifies, all 13 linear solves exhausted their 40-vector Krylov subspace at a relative residual of 0.26 against the 0.1 asked.

The missing piece is a preconditioner that holds the species rows. It is the open item that owns these cases.

Explanations excluded by measurement. These are, in order:

- *An outer boundary that traps H_2.* At cell 500 the diffusive face is exactly zero, because the carrier's ghost repeats the last value. Opening that face makes the refusing row worse, by factors 1.12 to 182: H_2 is heavier than the mean particle of the ionized wind, so the settling drift the open face admits points inward.
- *An inexact inner solve.* In each pass the element relaxation reached the fixed point of its own operator.
- *A two-state cycle.* The two-pass state difference is never small against the one-pass differences it spans.
- *A refusing cell that only appears to move.* It is fixed at cell 500 from pass 2 on in three of the four cases.

What does certify. The same network, with binary H/He diffusion at $K_{zz} = 10^9$, certifies from He/H = 0.55 to 9.7 (sections 4 and 5). What separates those states from the refused ones, beyond the measurements above, was not established. Three of the six certified later, by continuation under later binaries, with the carrier movement bound held (the next subsection); the account above is of the binaries it names.

Solved since, by continuation (2026-09-26 to 2026-09-29)

Most of the cases above, and the rungs between them, now certify (MEASURED; runs under `LHS1140b/models/.P1/s2_continuation/`, `LHS1140b/models/.P1/audit_F1/` and `LHS1140b/models/.P1/v5_test/`, `md/Update_EXHALE_stage3.md` sections 29.13, 30, 34 and 37). They are of the code after the changes named in the banner, so they are not comparable one to one with the older rows. All of them are published to the catalog as of 2026-09-29 and stand in the table of section 4 (notes d, e, g, h, i and r there): each with its own `run.log` by `models/publish_state.py` (the published pair is the solved pair byte for byte), its advection-corrected profiles and transit spectrum made by the no-step evaluate pass of the binary that solved it.

- **The well-mixed molecular family is the $K_{zz} = 10^9$ one with `He_diffusion off`**: its `base.inp` carries `Kzz_base 1.0e9`, which the carrier transport reads. A certified $K_{zz} = 10^9$ state with He/H set uniform (`map_state_to_grid.py --uniform`) certifies with diffusion off, and rungs in He/H then certify on the face-at-level grid from 2.13 down to 0.1: 2.13, 1.8, 1.4, 1.1, 0.85, 0.76, 0.68, 0.61, 0.55, 0.4, 0.366, 0.335, 0.317, 0.3, 0.26, 0.22, 0.19, 0.16, 0.14, 0.12 and 0.1, the rungs below 0.4 with the base contact direction held through each solve (section 30 of the log). The rung at 0.09 does not certify: the H_2 carrier row of the outermost cell ($r = 30\ R_p$) rises pass after pass (`md/TO_BE_DONE.md` item (m)). Starting the direct way from the catalog's refused state, the same case is refused again at pass 5.
- **The molecular $K_{zz} = 10^9$ family**: certified on the face-at-level grid from He/H = 1.45 down to 0.19, through 1.30, 1.223, 1.186, 1.168, 1.15, 1.12, 1.09, 1.075, 1.06, 1.03, 1.0, 0.97, 0.94, 0.91, 0.88, 0.85, 0.8, 0.75, 0.7, 0.65, 0.6, 0.55, 0.5, 0.45, 0.4, 0.374, 0.35, 0.324, 0.3, 0.279, 0.26 and 0.22. Below He/H $\approx$ 1.2 the rungs must be small (0.01 to 0.03 in He/H) and the carrier movement bound held (`md/TO_BE_DONE.md` item (g)). The rungs at 0.16 and at the log-midpoint 0.174 do not certify in 80 and 47 passes.
- **`molecular_photochem_gj1132_kzzprofile/HeH9`** certifies (41 passes) once the lower-ghost closure defect of section 29.9 of the log is fixed, and the photochemical He/H = 2.09 certifies as well. Both were reloaded with the carrier enthalpy flux and certified again, He/H = 9 in 24 passes and 2.09 in 47, and those states are the published ones.
- **With the carrier enthalpy flux** (section 34 of the log), eight of these states were reloaded and re-certified: $K_{zz} = 10^9$ at He/H = 0.55, 1.223, 1.30 and 1.45 (the last two by continuation from 1.223), well-mixed at 0.55 and 2.13, and the photochemical 2.09 and 9, in 6 to 47 passes; on the four compared with their parents (the $K_{zz} = 10^9$ and well-mixed 0.55 and the two photochemical ones) the term changes the temperature of the H_2 layer by −32 to +43 K and the wind mass flux by −0.2 to −0.9 per cent.
- **Under the discrete-equilibrium base boundary** (model v5, section 37 of the log), the well-mixed He/H = 0.55 state certified with the carrier enthalpy flux was reloaded and certified again in 4 passes (`EXHALE_3c22c63e.x`); that state is the published one.

- **Still unsolved**: $K_{zz} = 10^9$ below He/H = 0.19, well-mixed below 0.1, and the four atomic He/H = 9.7 models above (in the catalog's equation set; see the first item of the list above for the off-catalog chains with conduction, stopped 2026-09-30).

4. The models that are certified

146 models carry a certificate. The table lists each with the mass-loss rate and the He I 10830 equivalent width the catalog reports for it. On 2026-09-29 the continuation states of section 3 (the last subsection) were published: the two photochemical molecular rows, the molecular $K_{zz} = 10^9$ family from He/H = 0.19 to 1.45 (34 rows, 0.55 and 1.45 replacing their earlier states) and the well-mixed molecular family from He/H = 0.1 to 2.13 (21 rows). The family is therefore of mixed code, and the notes say which binary, boundary model and grid each row is of (notes d, e, g, h, i); five of them, and photochem_kzzprofile/HeH2.09, carry note r. The He/H = 9.7 rungs at 0.0675, 0.0662 and 0.065 of the fiducial XUV were certified on 2026-09-23 under `EXHALE_69233669.x`; the molecular $K_{zz} = 10^9$ rows at He/H = 1.52, 1.6, 1.8 and 2.13 were certified on 2026-09-26 under `EXHALE_7db5afe9.x` (note a) and 9.7 on 2026-09-20 (note c); the XUV continuation of the He/H = 2.13 branch (0.01 to 0.07) on 2026-09-21.

case	wind chemistry	He/H	XUV	K_{zz}	diff.	du	$\|\|R\|\|$	$\log_{10}\dot{M}$	EW
photochem_x0.10_kzzprofile/HeH9.7	atomic	9.7	0.10	profile	on	4.973E-10	1.022E-07	6.840	0.2674
photochem_x0.15_kzzprofile/HeH9.7	atomic	9.7	0.15	profile	on	4.692E-12	6.210E-08	7.020	0.4822
photochem_x0.20_kzzprofile/HeH9.7	atomic	9.7	0.20	profile	on	2.002E-12	6.628E-08	7.160	0.6711
photochem_x0.25_kzzprofile/HeH9.7	atomic	9.7	0.25	profile	on	1.900E-12	6.245E-08	7.260	0.8428
photochem_x0.30_kzzprofile/HeH9.7	atomic	9.7	0.30	profile	on	1.261E-10	3.190E-08	7.350	1.0097
photochem_x0.33_kzzprofile/HeH9.7	atomic	9.7	0.33	profile	on	1.644E-10	3.457E-08	7.390	1.0979
scalarCNO_kzz1e9/HeH2.13	atomic	2.13	1	1.0e9	on	1.739E-11	1.352E-08	7.870	1.3721
scalar_gj699_wellmixed/HeH0.042	atomic	0.042	1	none	off	2.983E-13	1.341E-07	8.570	0.9091
scalar_gj699_wellmixed/HeH0.046	atomic	0.046	1	none	off	1.827E-13	9.505E-08	8.570	0.9857
scalar_gj699_wellmixed/HeH0.050	atomic	0.050	1	none	off	1.587E-11	1.414E-07	8.570	1.0609
scalar_gj699_wellmixed/HeH0.083	atomic	0.083	1	none	off	7.470E-13	1.320E-07	8.570	1.6271
scalar_gj699_wellmixed/HeH1	atomic	1	1	none	off	5.765E-13	8.372E-08	8.550	4.5248
scalar_gj699_wellmixed/HeH1000	atomic	1000	1	none	off	8.965E-13	7.482E-09	8.420	2.8724
scalar_kzz0/HeH0.55	atomic	0.55	1	0.0	on	1.234E-10	1.707E-08	7.840	0.0305
scalar_kzz0/HeH2.6	atomic	2.6	1	0.0	on	4.029E-11	1.344E-08	7.860	0.9549
scalar_kzz0/HeH3.0	atomic	3.0	1	0.0	on	3.264E-11	2.215E-08	7.870	1.0798
scalar_kzz0/HeH3.5	atomic	3.5	1	0.0	on	2.429E-11	1.458E-08	7.870	1.2175
scalar_kzz0/HeH3.7	atomic	3.7	1	0.0	on	4.900E-11	1.776E-08	7.870	1.2698
scalar_kzz0/HeH3.9	atomic	3.9	1	0.0	on	4.557E-11	2.821E-08	7.870	1.3168
scalar_kzz1e10/HeH0.55	atomic	0.55	1	1.0e10	on	1.963E-11	1.870E-08	7.860	0.6463
scalar_kzz1e10/HeH1.06	atomic	1.06	1	1.0e10	on	1.994E-11	2.280E-08	7.860	1.0456
scalar_kzz1e10/HeH1.15	atomic	1.15	1	1.0e10	on	2.240E-11	1.463E-08	7.860	1.1031
scalar_kzz1e10/HeH1.29	atomic	1.29	1	1.0e10	on	2.711E-11	1.336E-08	7.860	1.1867
scalar_kzz1e11/HeH0.55	atomic	0.55	1	1.0e11	on	1.769E-11	2.546E-08	7.860	0.8065
scalar_kzz1e11/HeH0.795	atomic	0.795	1	1.0e11	on	5.628E-11	1.651E-08	7.860	1.0302
scalar_kzz1e11/HeH0.865	atomic	0.865	1	1.0e11	on	7.733E-11	1.951E-08	7.860	1.0866
scalar_kzz1e11/HeH0.93	atomic	0.93	1	1.0e11	on	1.864E-10	2.097E-08	7.860	1.1364
scalar_kzz1e5/HeH2.6	atomic	2.6	1	1.0e5	on	3.749E-11	1.394E-08	7.870	0.9631
scalar_kzz1e5/HeH3.0	atomic	3.0	1	1.0e5	on	2.952E-11	1.531E-08	7.870	1.0880
scalar_kzz1e5/HeH3.35	atomic	3.35	1	1.0e5	on	2.488E-11	1.509E-08	7.870	1.1862
scalar_kzz1e5/HeH3.64	atomic	3.64	1	1.0e5	on	4.781E-11	1.764E-08	7.870	1.2627
scalar_kzz1e5/HeH3.93	atomic	3.93	1	1.0e5	on	4.160E-11	1.552E-08	7.870	1.3314
scalar_kzz1e6/HeH0.55	atomic	0.55	1	1.0e6	on	1.895E-11	2.047E-08	7.840	0.0590
scalar_kzz1e6/HeH2.4	atomic	2.4	1	1.0e6	on	2.785E-11	1.570E-08	7.860	0.9465
scalar_kzz1e6/HeH2.8	atomic	2.8	1	1.0e6	on	4.856E-11	1.407E-08	7.870	1.0804
scalar_kzz1e6/HeH3.19	atomic	3.19	1	1.0e6	on	4.002E-11	2.368E-08	7.870	1.1948
scalar_kzz1e6/HeH3.46	atomic	3.46	1	1.0e6	on	3.570E-11	1.742E-08	7.870	1.2669
scalar_kzz1e6/HeH3.74	atomic	3.74	1	1.0e6	on	6.781E-11	2.390E-08	7.870	1.3388
scalar_kzz1e7/HeH0.55	atomic	0.55	1	1.0e7	on	1.289E-10	2.116E-08	7.850	0.1586
scalar_kzz1e7/HeH2.70	atomic	2.70	1	1.0e7	on	2.053E-12	1.497E-08	7.870	1.1996
scalar_kzz1e7/HeH2.94	atomic	2.94	1	1.0e7	on	1.639E-12	1.420E-08	7.870	1.2724
scalar_kzz1e7/HeH3.18	atomic	3.18	1	1.0e7	on	4.895E-11	3.913E-08	7.870	1.3265
scalar_kzz1e8/HeH0.55	atomic	0.55	1	1.0e8	on	5.543E-11	2.155E-08	7.850	0.3089
scalar_kzz1e8/HeH2.05	atomic	2.05	1	1.0e8	on	7.949E-11	1.428E-08	7.870	1.1513
scalar_kzz1e8/HeH2.23	atomic	2.23	1	1.0e8	on	7.229E-11	1.895E-08	7.870	1.2207
scalar_kzz1e8/HeH2.41	atomic	2.41	1	1.0e8	on	5.482E-11	1.624E-08	7.870	1.2826
scalar_kzz1e9/HeH0.55	atomic	0.55	1	1.0e9	on	6.072E-11	1.518E-08	7.850	0.4794
scalar_kzz1e9/HeH1.50	atomic	1.50	1	1.0e9	on	2.944E-11	1.646E-08	7.870	1.0996
scalar_kzz1e9/HeH1.60	atomic	1.60	1	1.0e9	on	4.581E-11	1.313E-08	7.870	1.1511
scalar_kzz1e9/HeH1.70	atomic	1.70	1	1.0e9	on	3.779E-11	2.343E-08	7.870	1.1969
scalar_kzz1e9/HeH2.13	atomic	2.13	1	1.0e9	on	2.462E-11	1.700E-08	7.870	1.3749
scalar_kzz1e9/HeH4.0	atomic	4.0	1	1.0e9	on	9.341E-13	1.470E-08	7.870	1.8229
scalar_kzz1e9/HeH9.7	atomic	9.7	1	1.0e9	on	5.192E-11	2.009E-08	7.860	2.0592
scalar_wellmixed/HeH0.083	atomic	0.083	1	none	off	9.755E-12	2.543E-08	7.840	0.3798
scalar_wellmixed/HeH0.40	atomic	0.40	1	none	off	1.442E-12	2.641E-08	7.850	1.1011
scalar_wellmixed/HeH0.42	atomic	0.42	1	none	off	3.329E-12	1.823E-08	7.850	1.1324
scalar_wellmixed/HeH0.44	atomic	0.44	1	none	off	2.623E-12	1.723E-08	7.850	1.1626

case	wind chemistry	He/H	XUV	K_{zz}	diff.	du	$\|\|R\|\|$	$\log_{10}\dot{M}$	EW
scalar_wellmixed/HeH0.55	atomic	0.55	1	none	off	1.618E-12	1.929E-08	7.850	1.3118
scalar_wellmixed/HeH1	atomic	1	1	none	off	4.004E-12	1.485E-08	7.860	1.6958
scalar_wellmixed/HeH10	atomic	10	1	none	off	1.205E-10	1.281E-08	7.850	1.9347
scalar_wellmixed/HeH100	atomic	100	1	none	off	1.684E-12	1.075E-08	7.860	1.9644
scalar_wellmixed/HeH1000	atomic	1000	1	none	off	1.744E-12	1.567E-08	7.850	1.9197
scalar_x0.015_kzz1e9/HeH2.13	atomic	2.13	0.015	1.0e9	on	1.775E-10	9.233E-07	5.970	0.0000
scalar_x0.01_kzz1e9/HeH2.13	atomic	2.13	0.01	1.0e9	on	2.220E-10	3.154E-07	5.790	0.0000
scalar_x0.02_kzz1e9/HeH2.13	atomic	2.13	0.02	1.0e9	on	8.304E-11	1.096E-07	6.100	0.0000
scalar_x0.03_kzz1e9/HeH2.13	atomic	2.13	0.03	1.0e9	on	4.353E-11	1.169E-07	6.280	0.0000
scalar_x0.05_kzz1e9/HeH2.13	atomic	2.13	0.05	1.0e9	on	6.354E-11	2.969E-07	6.510	0.0000
scalar_x0.065_kzz1e9/HeH9.7	atomic	9.7	0.065	1.0e9	on	2.649E-12	5.760E-08	6.610	0.0132
scalar_x0.0662_kzz1e9/HeH9.7	atomic	9.7	0.0662	1.0e9	on	2.534E-12	2.452E-08	6.620	0.0172
scalar_x0.0675_kzz1e9/HeH9.7	atomic	9.7	0.0675	1.0e9	on	3.317E-12	3.863E-08	6.630	0.0239
scalar_x0.07_kzz1e9/HeH2.13	atomic	2.13	0.07	1.0e9	on	5.100E-11	1.976E-07	6.660	0.0000
scalar_x0.07_kzz1e9/HeH9.7	atomic	9.7	0.07	1.0e9	on	6.972E-12	9.931E-08	6.640	0.0445
scalar_x0.10_kzz1e9/HeH2.13	atomic	2.13	0.10	1.0e9	on	1.464E-11	3.886E-07	6.820	0.0001
scalar_x0.10_kzz1e9/HeH9.7	atomic	9.7	0.10	1.0e9	on	4.424E-12	4.362E-07	6.810	0.2353
scalar_x0.15_kzz1e9/HeH2.13	atomic	2.13	0.15	1.0e9	on	9.116E-12	1.239E-07	7.000	0.0003
scalar_x0.15_kzz1e9/HeH9.7	atomic	9.7	0.15	1.0e9	on	8.205E-12	1.162E-07	7.000	0.4514
scalar_x0.20_kzz1e9/HeH2.13	atomic	2.13	0.20	1.0e9	on	1.072E-11	1.165E-07	7.130	0.0022
scalar_x0.20_kzz1e9/HeH9.7	atomic	9.7	0.20	1.0e9	on	5.463E-12	8.109E-08	7.130	0.6294
scalar_x0.25_kzz1e9/HeH2.13	atomic	2.13	0.25	1.0e9	on	5.191E-12	1.806E-07	7.240	0.0700
scalar_x0.25_kzz1e9/HeH9.7	atomic	9.7	0.25	1.0e9	on	1.677E-11	1.044E-07	7.230	0.7913
scalar_x0.30_kzz1e9/HeH2.13	atomic	2.13	0.30	1.0e9	on	3.218E-12	6.850E-08	7.320	0.1894
scalar_x0.30_kzz1e9/HeH9.7	atomic	9.7	0.30	1.0e9	on	5.587E-12	6.702E-08	7.310	0.9395
scalar_x0.33_kzz1e9/HeH2.13	atomic	2.13	0.33	1.0e9	on	2.761E-12	4.566E-08	7.360	0.2555
scalar_x0.33_kzz1e9/HeH9.7	atomic	9.7	0.33	1.0e9	on	3.376E-12	9.900E-08	7.360	1.0231
photochem_kzzprofile/HeH2.09[d,r]	molecular	2.09	1	profile	on	3.559E-12	1.590E-07	8.060	1.5009
photochem_kzzprofile/HeH9[d]	molecular	9	1	profile	on	4.885E-12	7.724E-08	7.990	1.8919
scalar_kzz1e9/HeH0.19[g,r]	molecular	0.19	1	1.0e9	on	1.173E-10	1.000E-08	7.340	0.0021
scalar_kzz1e9/HeH0.22[g]	molecular	0.22	1	1.0e9	on	6.063E-12	1.270E-08	7.380	0.0024
scalar_kzz1e9/HeH0.26[g]	molecular	0.26	1	1.0e9	on	5.135E-12	1.160E-08	7.440	0.0031
scalar_kzz1e9/HeH0.2793[g]	molecular	0.2793	1	1.0e9	on	3.765E-12	1.270E-08	7.460	0.0036
scalar_kzz1e9/HeH0.3[g]	molecular	0.3	1	1.0e9	on	3.026E-12	9.640E-09	7.490	0.0044
scalar_kzz1e9/HeH0.324[g]	molecular	0.324	1	1.0e9	on	3.561E-12	1.400E-08	7.520	0.0058
scalar_kzz1e9/HeH0.35[g]	molecular	0.35	1	1.0e9	on	2.078E-12	1.250E-08	7.550	0.0089
scalar_kzz1e9/HeH0.3742[g]	molecular	0.3742	1	1.0e9	on	3.547E-12	7.500E-09	7.580	0.0165
scalar_kzz1e9/HeH0.4[g]	molecular	0.4	1	1.0e9	on	1.990E-12	1.140E-08	7.600	0.0386
scalar_kzz1e9/HeH0.45[g]	molecular	0.45	1	1.0e9	on	4.610E-12	6.630E-09	7.640	0.0927
scalar_kzz1e9/HeH0.5[g]	molecular	0.5	1	1.0e9	on	1.998E-12	1.540E-08	7.670	0.1453
scalar_kzz1e9/HeH0.55[e]	molecular	0.55	1	1.0e9	on	1.256E-12	9.220E-09	7.700	0.1972
scalar_kzz1e9/HeH0.6[g]	molecular	0.6	1	1.0e9	on	1.964E-12	1.360E-08	7.730	0.2580
scalar_kzz1e9/HeH0.65[g]	molecular	0.65	1	1.0e9	on	2.410E-12	7.420E-09	7.750	0.3144
scalar_kzz1e9/HeH0.7[g]	molecular	0.7	1	1.0e9	on	1.910E-12	6.100E-09	7.770	0.3669
scalar_kzz1e9/HeH0.75[g,r]	molecular	0.75	1	1.0e9	on	2.413E-12	1.010E-08	7.780	0.4152
scalar_kzz1e9/HeH0.8[g]	molecular	0.8	1	1.0e9	on	1.962E-12	1.080E-08	7.790	0.4603
scalar_kzz1e9/HeH0.85[g,r]	molecular	0.85	1	1.0e9	on	1.963E-12	1.410E-08	7.810	0.5029
scalar_kzz1e9/HeH0.88[g]	molecular	0.88	1	1.0e9	on	2.304E-12	1.100E-08	7.810	0.5274
scalar_kzz1e9/HeH0.91[g]	molecular	0.91	1	1.0e9	on	1.469E-12	1.060E-08	7.820	0.5513
scalar_kzz1e9/HeH0.94[g]	molecular	0.94	1	1.0e9	on	2.029E-12	8.240E-09	7.830	0.5743
scalar_kzz1e9/HeH0.97[g]	molecular	0.97	1	1.0e9	on	2.799E-12	9.730E-09	7.830	0.5968
scalar_kzz1e9/HeH1.0[g]	molecular	1.0	1	1.0e9	on	2.390E-12	1.300E-08	7.840	0.6187
scalar_kzz1e9/HeH1.03[g]	molecular	1.03	1	1.0e9	on	2.496E-12	1.480E-08	7.840	0.6401
scalar_kzz1e9/HeH1.06[g,r]	molecular	1.06	1	1.0e9	on	2.301E-12	1.480E-08	7.850	0.6610
scalar_kzz1e9/HeH1.075[g]	molecular	1.075	1	1.0e9	on	2.553E-12	1.500E-08	7.850	0.6712
scalar_kzz1e9/HeH1.09[g]	molecular	1.09	1	1.0e9	on	2.090E-12	1.260E-08	7.850	0.6814
scalar_kzz1e9/HeH1.12[g]	molecular	1.12	1	1.0e9	on	8.226E-13	8.120E-09	7.860	0.7013
scalar_kzz1e9/HeH1.15[g]	molecular	1.15	1	1.0e9	on	9.272E-13	9.490E-09	7.860	0.7207
scalar_kzz1e9/HeH1.168[g]	molecular	1.168	1	1.0e9	on	9.653E-13	1.070E-08	7.860	0.7322
scalar_kzz1e9/HeH1.186[g]	molecular	1.186	1	1.0e9	on	9.740E-13	7.990E-09	7.860	0.7435
scalar_kzz1e9/HeH1.223[d]	molecular	1.223	1	1.0e9	on	1.834E-12	1.250E-08	7.870	0.7621
scalar_kzz1e9/HeH1.30[d]	molecular	1.30	1	1.0e9	on	1.768E-12	1.960E-08	7.880	0.8076
scalar_kzz1e9/HeH1.45[d]	molecular	1.45	1	1.0e9	on	3.437E-12	3.220E-08	7.890	0.8896
scalar_kzz1e9/HeH1.52[a]	molecular	1.52	1	1.0e9	on	2.068E-12	7.337E-09	7.890	1.0149
scalar_kzz1e9/HeH1.6[a]	molecular	1.6	1	1.0e9	on	1.441E-12	1.167E-08	7.900	1.0577
scalar_kzz1e9/HeH1.8[a]	molecular	1.8	1	1.0e9	on	4.134E-12	8.818E-09	7.910	1.1550
scalar_kzz1e9/HeH2.13[a]	molecular	2.13	1	1.0e9	on	3.364E-11	1.266E-08	7.930	1.2902
scalar_kzz1e9/HeH9.7[c]	molecular	9.7	1	1.0e9	on	3.491E-11	2.580E-08	7.940	2.4060
scalar_wellmixed/HeH0.1[g]	molecular	0.1	1	none	off	9.526E-11	8.570E-09	7.150	0.1167
scalar_wellmixed/HeH0.12[g]	molecular	0.12	1	none	off	8.428E-12	1.900E-08	7.190	0.1426
scalar_wellmixed/HeH0.14[g]	molecular	0.14	1	none	off	6.625E-12	1.320E-08	7.220	0.1677
scalar_wellmixed/HeH0.16[g]	molecular	0.16	1	none	off	1.779E-10	2.280E-08	7.260	0.1933
scalar_wellmixed/HeH0.19[i]	molecular	0.19	1	none	off	1.939E-10	1.110E-08	7.310	0.2280
scalar_wellmixed/HeH0.22[g]	molecular	0.22	1	none	off	7.416E-12	8.620E-09	7.360	0.2753
scalar_wellmixed/HeH0.26[g]	molecular	0.26	1	none	off	4.177E-12	8.550E-09	7.420	0.3351
scalar_wellmixed/HeH0.3[g]	molecular	0.3	1	none	off	3.016E-12	1.540E-08	7.480	0.4025
scalar_wellmixed/HeH0.3169[g]	molecular	0.3169	1	none	off	3.552E-12	1.140E-08	7.490	0.4340
scalar_wellmixed/HeH0.3347[g]	molecular	0.3347	1	none	off	2.630E-12	5.030E-09	7.510	0.4692
scalar_wellmixed/HeH0.3659[g]	molecular	0.3659	1	none	off	2.831E-12	1.390E-08	7.540	0.5247
scalar_wellmixed/HeH0.4[g]	molecular	0.4	1	none	off	2.078E-12	1.400E-08	7.570	0.5754
scalar_wellmixed/HeH0.55[e]	molecular	0.55	1	none	off	9.797E-12	1.282E-08	7.660	0.7562

case	wind chemistry	He/H	XUV	K_{zz}	diff.	du	$\|\|R\|\|$	$\log_{10}\dot{M}$	EW
scalar_wellmixed/HeH0.6116[g]	molecular	0.6116	1	none	off	1.511E-12	9.050E-09	7.690	0.8268
scalar_wellmixed/HeH0.68[h]	molecular	0.68	1	none	off	1.328E-12	1.300E-08	7.730	0.9487
scalar_wellmixed/HeH0.7603[h]	molecular	0.7603	1	none	off	1.066E-12	7.880E-09	7.750	1.0274
scalar_wellmixed/HeH0.85[h]	molecular	0.85	1	none	off	7.789E-13	1.130E-08	7.780	1.1074
scalar_wellmixed/HeH1.1[h]	molecular	1.1	1	none	off	1.551E-12	9.940E-09	7.830	1.2915
scalar_wellmixed/HeH1.4[h]	molecular	1.4	1	none	off	2.509E-10	9.480E-09	7.870	1.4543
scalar_wellmixed/HeH1.8[h]	molecular	1.8	1	none	off	3.991E-11	9.140E-09	7.900	1.6043
scalar_wellmixed/HeH2.13[d,r]	molecular	2.13	1	none	off	1.139E-11	3.070E-08	7.910	1.5845

[a] Certified on 2026-09-26 under the binary of record, `EXHALE_7db5afe9.x`: the only rows of this table under the corrected atomic physics (section 5). [b] Certified on 2026-09-25 under `EXHALE_b059a4be.x`, after the second of the five rounds of corrections: neither the physics before them nor the corrected one. [c] The certified generation is the 2026-09-20 evaluation of the state written on 2026-09-19. A re-solve made later on 2026-09-20 did not certify (the H_2 carrier row 5.1×10^{-4} at cell 262 after 40 passes); its `not_solved.md` stays beside the case and describes that re-solve, not the state listed here. The du and $\|R\|$ of this row are that re-solve's, as the rule of the column paragraph above gives them. [d] Certified under `EXHALE_6dd54955.x`, md5 `6dd549556910351a7f857708b4853a69` (the binary of record), with the base boundary model v4 (`. . . _face_at_level_v4`) on the face-at-level grid and the enthalpy flux of the transported carriers on. The two photochemical rows were solved in `.P1/audit_F1/photo9_v4_F1` and `.P1/audit_F1/photo209_v5_F1` (generations `g0006_20260929T054534Z_c399da20` and `g0001_20260929T055054Z_689dcef6`); the others are $K_{zz} = 10^9$ continuation rungs of `.P1/s2_continuation/` and `.P1/audit_F1/`. [e] Certified under `EXHALE_3c22c63e.x`, md5 `3c22c63e787a017e7bbd88186cbc8b09`, the Intel build of the source with the discrete-equilibrium base boundary, model v5 (`. . . _discrete_equilibrium_v5`), on the face-at-level grid and with the enthalpy flux of the transported carriers on; its source list is `.P1/v5_test/BINARY_MANIFEST_3c22c63e787a.txt`. Solved in `.P1/v5_test/wm4_0.55_v5` and `.P1/v5_test/kzz8_0.55_v5`. [g] A continuation rung certified on 2026-09-27 to 2026-09-29 on the face-at-level grid with the base boundary model v4 and without the enthalpy flux of the transported carriers (the term did not yet exist), under `EXHALE_b037d586.x`, `EXHALE_8dbc44fb.x`, `EXHALE_a7255a03.x` or `EXHALE_2dd9a865.x`, as the case's generation manifest and `REPRODUCE.md` name. [h] A well-mixed continuation rung certified on 2026-09-26 under `EXHALE_7db5afe9.x` on the grid before the face-at-level change (boundary model v3) and without the carrier enthalpy flux. [i] Certified under `EXHALE_b029a1e8.x`, the gfortran build used to measure the carrier enthalpy flux (term on, boundary v4); no binary manifest was written for it. [r] The no-step evaluate pass of the same binary, which made the advection-corrected profiles, does not reproduce the stationary claim of the published state; the index keeps the solve's certificate, and the refusing evaluation was not published (published, it would take the certified reference away, `LHS1140b/MODELS.md` section 9.8). On photochem_kzzprofile/HeH2.09 the reloaded state's cell-1 mass row reads 1.25×10^{-8} against its tolerance 6.6×10^{-9}, where the solve's own certificate read 2.77×10^{-9} (MEASURED); the reload re-derives the two lower ghost rows. Why a reload moves the base cell's mass row past its tolerance is open. Every row published on 2026-09-29 is the solved pair byte for byte, with its own `run.log`; the scripts are in `.P1/publish_rows/`. Every unmarked row is of the physics before the corrections.

5. The molecular $K_{zz} = 10^9$ family across the observed line

STALE (2026-10-07). Every molecular state run with `Molecular carrier transport: True` and the oxygen chemistry off carries a code defect found on 2026-10-07: the transported H_2 row kept the loss $O + H_2 \rightarrow OH + H$ without production, reverse or reaction heat (`md/AN_matching_surface_diagnosis_20261007.md`). The numbers of such states below are records of those runs, not results, until recomputed.

What changed, and on which binary. The atomic data of EXHALE were checked against their original sources on 2026-09-24 to 2026-09-26 (`md/Update_EXHALE_stage3.md` sections 17 to 26). The corrections that move this line most are the electron-impact de-excitation $2^3S \rightarrow 1^1S$ (the detailed-balance reverse of the excitation, omitted by the metastable network before), the He I collision strengths of Bray et al. (2000) in place of an extrapolated fit, the case-A/B escape of the H ground captures, and the $He^+ + H$ charge exchange, whose non-radiative channel the Kingdon & Ferland (1996) fit had overstated by a factor of about 200 at 10^4 K. Every number of this section is from states certified under that physics (`EXHALE_59c6be36.x`; the binary of record of the catalog rows, `EXHALE_7db5afe9.x`, differs from it only in refusing, rather than stopping at, an unclosed lower ghost of a trial state, and gives byte-identical results on every run that does not meet it). The values of this section before the audit are in sections 11 to 15 of the update log. The He/H = 0.55 and 9.7 molecular rows of the catalog were not re-solved and are of the physics before the audit; the 1.45 row is of the second round of corrections (`EXHALE_b059a4be.x`, EW 0.9387). Under the

third round (`EXHALE_50aa18d7.x`) neither its re-solve nor its rung from 1.52 certified before its time ceiling, and it was not attempted under the binary of record. (Both the 0.55 and the 1.45 case have published a later generation since, on 2026-09-29, from the continuation (EW 0.1972 and 0.8896, section 4); this section does not use them, and neither changes the pair of rungs that straddles the observed width.)

The rungs. He/H = 1.8 and 2.13 were re-solved from their certified states of the previous physics, and 1.6 and 1.52 as rungs from 1.8 (the rung to 1.45, made under the third round, did not certify: the H_2 carrier row held at 10^{-1} through four passes). Each certified state was published into the catalog through `run_case.sh`, and every re-measurement reproduced its claim. MEASURED:

He/H	passes	EW [% Å]	$\log_{10}\dot{M}$ [g s^{-1}]
1.52	24	1.0149	7.89
1.6	25	1.0577	7.90
1.8	23	1.1550	7.91
2.13	21	1.2902	7.93
observed		1.108 ± 0.030	

The certified rungs 1.6 and 1.8 straddle the observed EW, and the log-log chord crosses 1.108 at He/H = 1.703 on the catalog grid (DERIVED); before the audit the crossing was 1.464. The profiles move monotonically with He/H (Fig. 3).

The grid of the crossing. The three straddling compositions were solved again on 1000 and 2000 cells (the base block and every other spacing halved at each step, the outer radius unchanged); all nine states certify, and every evaluation. MEASURED, with the continuum DERIVED by Richardson extrapolation at the measured order:

	500 cells	1000 cells	2000 cells	continuum
EW, He/H = 1.6	1.0577	1.0297	1.0150	0.999
EW, He/H = 1.8	1.1550	1.1249	1.1094	1.093
EW, He/H = 2.13	1.2902	1.2571	1.2403	1.223
crossing of 1.108, EW	1.703	1.764	1.797	1.838
crossing of 1.108, broadened EW	1.781	1.851	1.890	1.933

The EW converges at first order in the cell width (successive-difference ratios 1.91 to 1.96, $p = 0.93$ to 0.97), as it did before the audit; a formally second-order scheme converging at first order in the EW points to a first-order error in the line-forming region whose source is not established. The continuum crossings are interpolated between the continuum points of 1.8 and 2.13. The 60 R_p domain factor measured before the audit (the EW lower by 1.7 per cent) was not measured again and is not applied.

How the grid is used (decision of 2026-09-24). The catalog keeps its 500-cell grid. A composition read from it is quoted as the continuum value of the table above, and the difference between the 500-cell and the continuum value is the grid uncertainty of every catalog number: from the EW, He/H = 1.84 (1.70 on the catalog grid, 0.033 dex lower, the 500-cell EW being about 6 per cent high); from the broadened line, about 1.93 (1.78 on the catalog grid).

The width of the line. The EW is matched, the width is not: the unbroadened red line of He/H = 1.8 is 4.18 per cent deep and 0.259 Å wide, against the fitted 1.24 per cent and the measured 0.841 Å. As in the p-winds comparison memo, the excess absorption of the instrument-convolved spectrum is convolved with a Gaussian of FWHM f in velocity, its depth, blue depth, ratio and red-pair FWHM are read with the three-Gaussian extractor used on the data, and f is matched where the FWHM reaches the measured 0.841 Å (the memo's atomic well-mixed He/H = 0.40 state is reproduced, $f = 22.22$ km s^{-1}; `.P1/connectivity/broaden_molecular_final.py`). MEASURED:

He/H	cells	EW, $f = 0$	matched f [km s^{-1}]	red [%]	blue [%]	red/blue	EW at f
1.52	500	1.0149	22.19	1.133	0.148	7.64	0.9813
1.6	500	1.0577	22.18	1.181	0.155	7.63	1.0227
1.8	500	1.1550	22.15	1.289	0.170	7.60	1.1167
2.13	500	1.2902	22.12	1.440	0.190	7.57	1.2474
1.6	1000	1.0297	22.17	1.149	0.151	7.64	0.9955
1.8	1000	1.1249	22.14	1.255	0.165	7.61	1.0876
2.13	1000	1.2571	22.11	1.403	0.185	7.58	1.2155
1.6	2000	1.0150	22.16	1.133	0.148	7.64	0.9813
1.8	2000	1.1094	22.14	1.238	0.163	7.62	1.0726
2.13	2000	1.2403	22.10	1.384	0.182	7.59	1.1992
observed		1.108 ± 0.030		$1.24^{+0.22}_{-0.23}$	$0.25^{+0.14}_{-0.12}$	$6.7^{+12.7}_{-3.1}$	1.108 ± 0.030

Three readings of the table.

- *The kernel is set by the velocity field, not by the composition or the grid:* 22.10 to 22.19 km s^{-1} on every state, the value the memo found for the atomic families and the one found here before the audit. At it the He/H = 1.8 line has depth, blue depth and ratio inside the observed 1σ (Fig. 2).
- *The broadened EW moves the crossing* up by about 0.08 in He/H: the Gaussian wings carry absorption out of the red-pair window, lowering the EW by 3.3 per cent at the matched kernel.
- *As turbulence the kernel is supersonic.* On the advection-corrected profiles of He/H = 1.6 and 1.8, weighted over the line-forming region by $n(2\,^3S)\,r$, $T = 2650$ and 2756 K, $\mu = 1.94$ and 2.03 and $c_s = 4.33$ km s^{-1}; the matched $\sigma = 9.4$ km s^{-1} is $\mathcal{M} = 2.2$ as a line-of-sight dispersion, 3.1 as $\sqrt{2}\sigma$ and 3.8 as an isotropic speed.

How the kernel is used (decision of 2026-09-24). The added Gaussian is an empirical parameter of the comparison, fixed by the measured width alone, and not a physical term of the model. Every composition this catalog reads from the full line is conditional on it: with the kernel, He/H = 1.78 on the 500-cell catalog grid and about 1.93 in the continuum limit of the grid; without it, from the EW alone, 1.70 and 1.84.

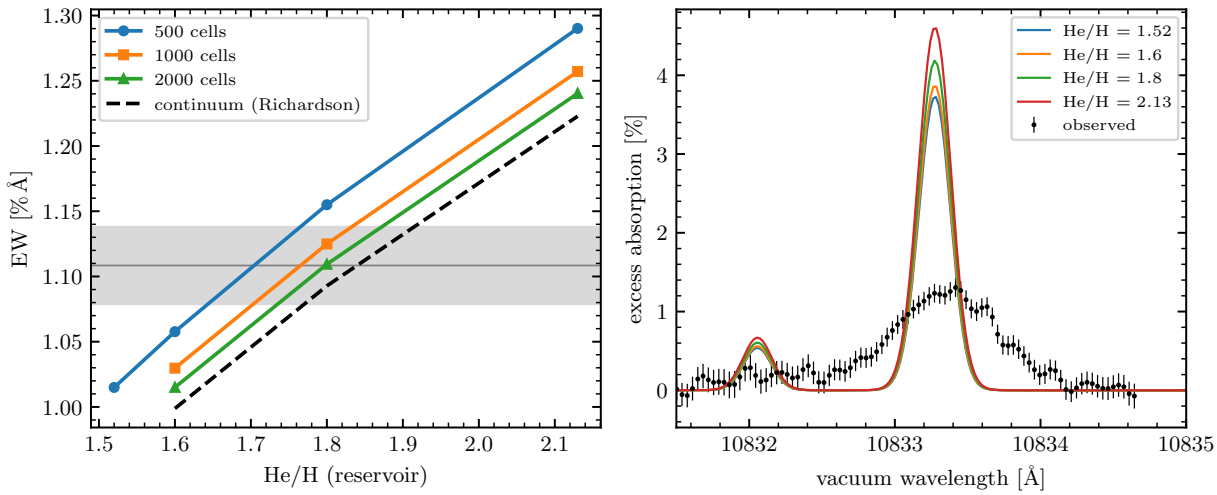


Figure 1: Left: the red-pair EW of the certified molecular states against the reservoir He/H on 500, 1000 and 2000 cells and in the continuum limit, with the observed 1.108 ± 0.030 % Å (grey). Right: the unbroadened 500-cell spectra against the released one.

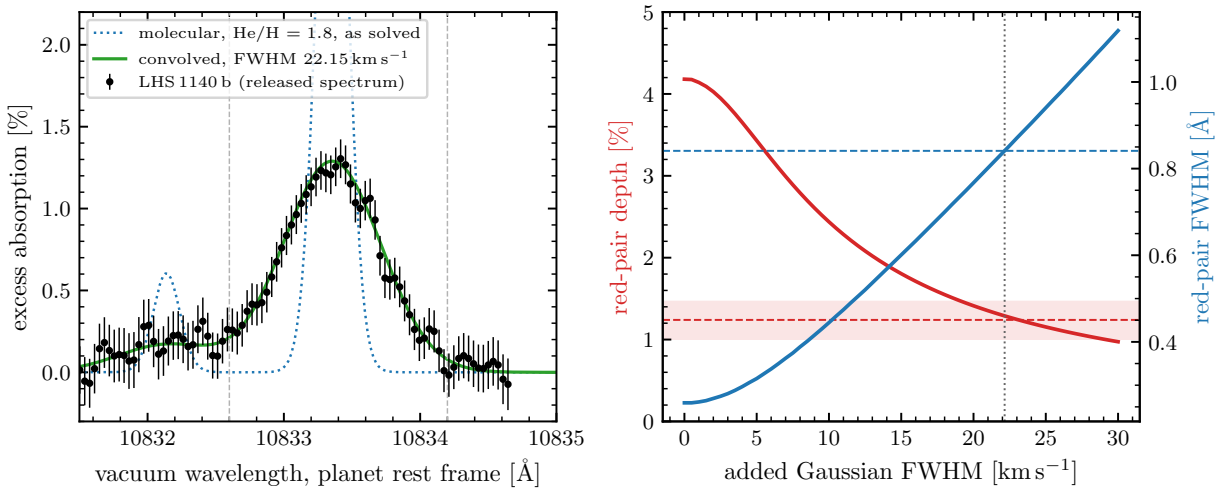


Figure 2: Left: the molecular He/H = 1.8 profile as solved (dotted) and convolved with a 22.15 km s^{-1} FWHM Gaussian (solid), against the released spectrum; both model curves are shifted by the memo's $v_{\rm wind}$ = +2.26 km s^{-1} for display (every EW is from the unshifted arrays), and the dashed verticals mark the EW window. Right: the red-pair depth (red, left axis) and FWHM (blue, right axis) as the added Gaussian is scanned; dashed lines are the observed values, the band the 1σ interval of the fitted depth, and the dotted vertical the matched kernel.

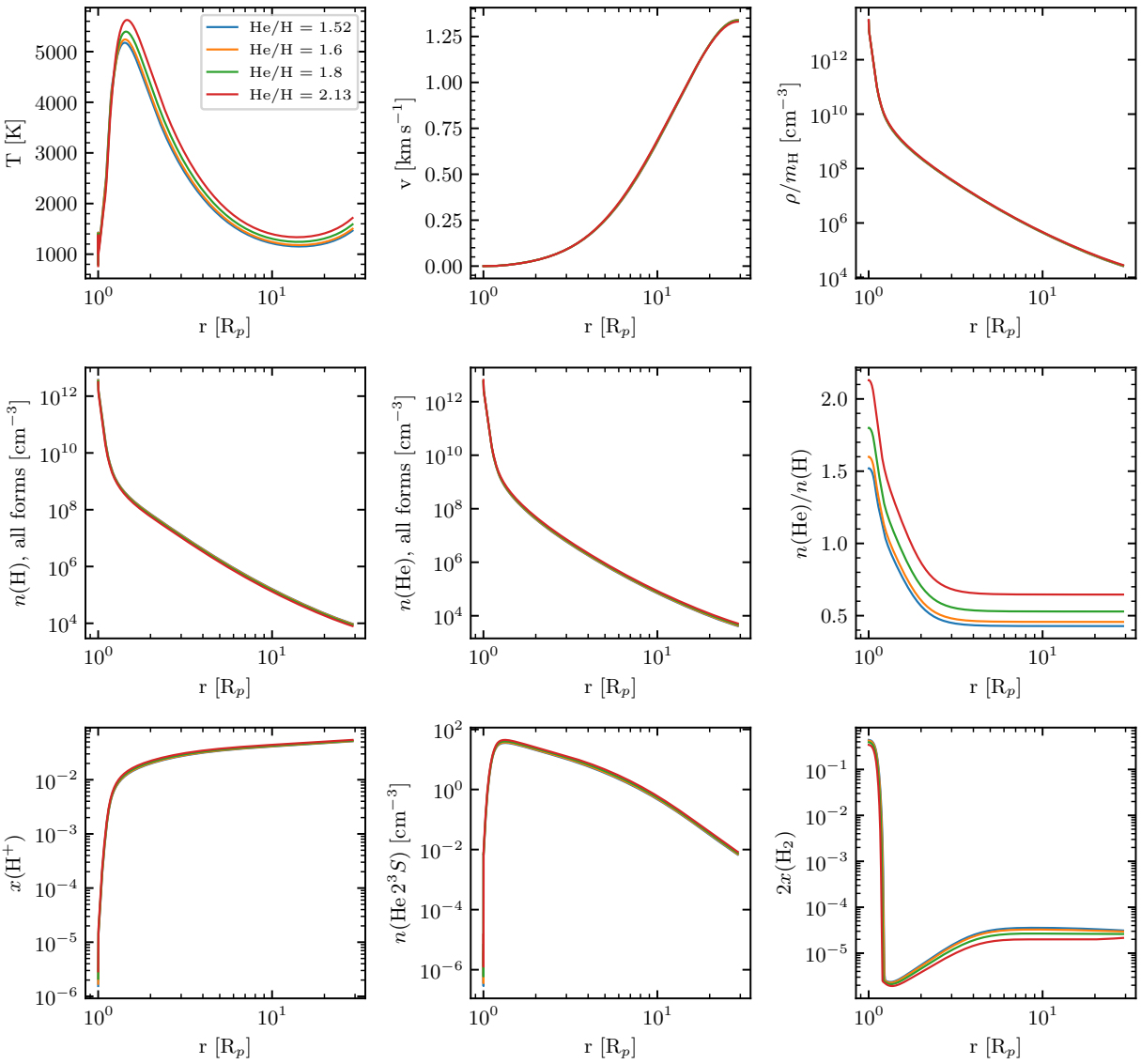


Figure 3: The certified molecular $K_{zz} = 10^9$ states He/H = 1.52 to 2.13 under the corrected atomic physics (advection-corrected profiles): temperature, velocity, $\rho/m_{\rm H}$, the number densities of all hydrogen nuclei n(H) and all helium nuclei n(He) (every ionization stage, level and molecule counted: H I, H II, $2\,H_2$, $2\,H_2^+$, $3\,H_3^+$ and HeH^+ for hydrogen; He I, He II, He III and HeH^+ for helium), their ratio n(He)/n(H), the H II fraction, $n(\mathrm{He}\,2^3S)$ and the H_2 fraction against radius.

6. The He 10830 EW with the full equation set (outside the catalog)

STALE (2026-10-07). Every molecular state run with `Molecular carrier transport: True` and the oxygen chemistry off carries a code defect found on 2026-10-07: the transported H_2 row kept the loss $O + H_2 \rightarrow OH + H$ without production, reverse or reaction heat (`md/AN_matching_surface_diagnosis_20261007.md`). The numbers of such states below are records of those runs, not results, until recomputed.

What is solved. None of the states of this section is published to the catalog; they are the runs of the EW plans `md/PLAN_20261003_lhs1140b_ew.md` and `md/PLAN_20261004_lhs1140b_heh_high_and_xuv.md`, under `LHS1140b/models/.P2/ew_fit/` (log sections 63 to 102). READ from the input template and the `closure.json` of each case (`route_heh_high/HeH10.0/` shown here; the other cases differ only in He/H, the spectrum, the binary and the column options named below): the molecular network with carrier transport, binary H/He diffusion, `Conduction: True`, `Ionization transport: True`, `He 2^3S transport: True`, `Secondary_ionization: Immediate`, `Well balanced: True`, `Domain mode: Spherical` to 30 R_p, and a Photochem column below the 1 microbar matching level handed over as a profile (`Lower atmosphere profile:`) with $K_{zz} = 10^9$ cm^2 s^{-1} in the column, its energy treatment `nlte` and the conducted heat flux of the wind in its energy balance. Column and wind are iterated by `src/utils/element_flux_closure.py` with `energy_coupling` on (`energy_coupling_omega` 0.5, `T_match_tol` 6 K; elemental-flux tolerance 0.05, from the log). The EWs are READ from the `EW.txt` that `.P1/fiducial_conduction/ew_of_state.sh` wrote with `EW_NONTHERMAL_KERNEL=width_match` for the last closure iterate (`written_state certified=T` in every file read): *EW* is the unbroadened `EW=` of the solution product, *EW wm* the width-matched `EW_kernel=` and f its `fwhm_kms=`. The kernel stands for a velocity field the 1-D model does not contain; it is not a computed physics term. Observed: 1.1084 ± 0.0289 % Å.

The EW against deep He/H on the GJ 1132 spectrum. READ from `phase5/lowHeH_numbers.txt` (He/H ≤ 1.8) and `route_heh_high/route_heh_high_numbers.txt` (He/H ≥ 2.09), the tables the comparison scripts measured from the case directories; the EW columns were checked against the `EW.txt` of He/H = 0.65, 2.09 and 7000. The column is the band transfer of the Photochem column balance (`--column-band-transfer`). Rows marked † were solved by `EXHALE_99a76cb0.x` (1.8: `EXHALE_8cc7b546.x`) before the hydrogen carrier split of the escape boundary condition was made self-consistent (log section 95); the others by `EXHALE_c37b3671.x` (`exhale_bin` of `closure.json`). Every listed closure converged and every wind state is certified.

deep He/H	closure k	column transfer	$T_{\rm match}$ [K]	$\log_{10}\dot{M}$	EW	EW wm	f [km s^{-1}]
0.65†	2	cooling to space	412.59	8.1325	0.8822	0.8529	22.24
0.8†	1	cooling to space	410.53	8.0930	0.8734	0.8444	22.21
1.0†	1	cooling to space	399.39	8.0531	0.8655	0.8368	22.17
1.2†	1	cooling to space	386.37	8.0218	0.8593	0.8308	22.14
1.5†	1	cooling to space	375.98	7.9770	0.8327	0.8050	22.09
1.8†	3	thin	359.55	7.9359	0.7975	0.7710	22.05
2.09	1	cooling to space	350.71	7.8864	0.7353	0.7109	22.02
3.0	1	cooling to space	332.83	7.8255	0.6993	0.6761	21.92
4.0	1	cooling to space	326.05	7.8007	0.7054	0.6820	21.85
5.0	1	cooling to space	321.73	7.8012	0.7399	0.7154	21.80
7.0	1	cooling to space	328.84	7.8135	0.8058	0.7791	21.73
10	1	cooling to space	334.06	7.8136	0.8398	0.8121	21.67
14	1	cooling to space	335.13	7.8067	0.8493	0.8212	21.62
20	1	cooling to space	343.02	7.7936	0.8416	0.8137	21.57
30	1	cooling to space	357.62	7.7749	0.8212	0.7941	21.53
40	2	cooling to space	371.07	7.7661	0.8137	0.7868	21.51
60	2	cooling to space	394.30	7.7599	0.8130	0.7862	21.48
80	2	cooling to space	412.25	7.7579	0.8160	0.7891	21.47
100	2	cooling to space	428.94	7.7576	0.8201	0.7931	21.46
200	1	cooling to space	484.29	7.7596	0.8362	0.8086	21.44
400	1	cooling to space	546.76	7.7657	0.8573	0.8291	21.44
700	1	cooling to space	602.09	7.7731	0.8779	0.8491	21.44
1000	1	cooling to space	635.41	7.7781	0.8913	0.8621	21.44
1400	1	cooling to space	667.00	7.7833	0.9047	0.8750	21.44
2000	1	cooling to space	703.55	7.7897	0.9205	0.8904	21.44
3000	1	cooling to space	748.33	7.7979	0.9406	0.9098	21.44
4000	1	cooling to space	779.89	7.8038	0.9550	0.9238	21.45
6000	1	cooling to space	816.20	7.8108	0.9725	0.9407	21.45
7000	1	cooling to space	826.69	7.8131	0.9781	0.9461	21.45
8000	1	cooling to space	837.95	7.8153	0.9836	0.9515	21.45
10^4	1	cooling to space	855.53	7.8188	0.9923	0.9600	21.45
observed					1.1084 ± 0.0289	1.1084 ± 0.0289	

On this spectrum no deep He/H from 0.65 to 10^4 reaches the observed EW (Fig. 4): the width-matched EW has a minimum of 0.676 at He/H = 3, rises to 0.853 at 0.65 and to 0.960 at 10^4. The closure at He/H = 0.5 did not converge (diagnostic EW wm 0.8482, the open circle of the figure; READ, `phase5/lowHeH_numbers.txt`). The phase-3 reference at He/H = 2.09 with the optically thin column (`phase3/he23s_transport/solve/`, `EXHALE_4c0e8603.x`) gave EW 0.7334 and EW wm 0.7091 (READ, `phase5/lowHeH_numbers.txt`); the 2.09 row above is its successor on the cooling-to-space column and the binary of log section 94. The width-matching kernel is 21.4 to 22.2 km s^{-1} on every state and lowers the window EW by about 3.3 per cent.

Numerical and observational checks at He/H = 0.65 (log section 92, taken from the log, `phase6/NOTES.md`): halving every spacing (1000 cells, certified) lowers EW wm by 0.0073, a further halving (diagnostic) by 0.0040, a Richardson continuum 0.016 below the 500-cell value; a tighter wind tolerance changes nothing, a tighter closure +0.0004, the domain to L1 +0.0020; the Roche-consistent geometry gives 0.8205 against the spherical 0.8529. Every positive item together reaches at most 0.870, 8.2σ below the observation.

The EUV that matches the observed EW. Holding the bolometric flux, the GJ 1132 spectrum shortward of 911.75 Å is scaled by s (a sensitivity construction, not a stellar model; log sections 98, 99 and 102). READ from `route_xuv_scaled/euv_match_vs_heh.txt`: s at which the width-matched EW interpolated in $\ln s$ reaches 1.1084 (in brackets the values for 1.1084 ∓ 0.0289), and $F_{\rm EUV}$(100–911 Å) at the orbit in erg cm^{-2} s^{-1}; GJ 1132 itself gives 10.05 (READ, `route_xuv_scaled/xuv_scaled_numbers.txt`, $s = 1$). Every He/H has a converged state within 0.0058 of the observation (the last column); a value marked * lies outside the bracket of converged states and is extrapolated along the end slope, as the file marks it; the He/H = 0.1 closure uses `--column-band-transfer cooling_to_space_exchange` and 24 top-pressure passes, the others `cooling_to_space` (`closure.json`, READ).

deep He/H	s	$F_{\rm EUV}$	nearest converged state: s, EW wm − observed
0.1	2.532* [2.430, 2.636*]	25.45 [24.42, 26.50]	2.53, −0.0005
0.2	1.656 [1.599, 1.715*]	16.65 [16.07, 17.23]	1.66, +0.0020
0.3	1.448 [1.402, 1.494*]	14.56 [14.09, 15.02]	1.45, +0.0012
0.45	1.339 [1.300, 1.379]	13.46 [13.07, 13.86]	1.34, +0.0006
0.65	1.300 [1.264, 1.337]	13.07 [12.71, 13.43]	1.30, 0.0000
1.2	1.280 [1.249, 1.311]	12.86 [12.55, 13.18]	1.28, +0.0004
2.09	1.361 [1.333, 1.390]	13.68 [13.40, 13.97]	1.36, −0.0013
5	1.336 [1.309, 1.363]	13.43 [13.16, 13.70]	1.34, +0.0045
14	1.271 [1.241, 1.302]	12.77 [12.48, 13.08]	1.27, −0.0008
100	1.331 [1.298, 1.365]	13.38 [13.04, 13.72]	1.33, −0.0008
1000	1.253 [1.223*, 1.286]	12.60 [12.29, 12.92]	1.25, −0.0031
10^4	1.143 [1.115, 1.173]	11.49 [11.20, 11.79]	1.14, −0.0031

The observed line needs 1.14 to 2.5 times the EUV of GJ 1132, nearly flat at 12.8 to 13.7 erg cm^{-2} s^{-1} over He/H 0.45 to 100 (Fig. 5). The states at He/H = 0.65 and $s = 1.3$ (EW wm 1.1084, $f = 22.20$ km s^{-1}) and at 2.09 and $s = 1.36$ (1.1071, 21.97 km s^{-1}) were checked in their `EW.txt`.

The GJ 699 spectrum. READ from `route_xuv/route_xuv_numbers.txt` (log section 95): on the GJ 699 proxy spectrum at He/H = 2.09 the closure was not iterated past $k = 0$ (EW wm 3.698, $f = 21.52$); with GJ 699 shortward of 2600 Å and GJ 1132 above ("split"), converged at He/H = 0.65, 2.09, 5 and 10, EW wm 3.707, 3.388, 3.427 and 3.328 (f 21.3 to 21.6 km s^{-1}), three times the observation; at He/H = 0.65 the unsplit GJ 699 closure was refused at $k = 0$ by the exchange bound of the cooling-to-space column transfer. The $F_{\rm EUV}$ of GJ 699 is 70.2 (log section 102).

7. What the flux-spread criterion misses, on one certified case

The question the columns above raise is what a state that meets $du < 10^{-3}$ but fails this code's certification differs from the certified stationary solution BY. Measured on the atomic `scalar_kzz1e9/HeH2.13`, which is certified, against a state of the same case stopped after one outer pass of the same route:

	certified stationary	flux converged, refused
du, the run's own gate	2.46E-11	1.83E-12
$\|\|R\|\|$, re-evaluated	1.61E-08	4.97E-08
the three hydrodynamic rows	within	within
elemental transport He/H, $r \geq 1.2$	within	5.21E-05 above 1.0E-05 at $r = 1.695$
verdict	CERTIFIED	NOT CERTIFIED, one entry

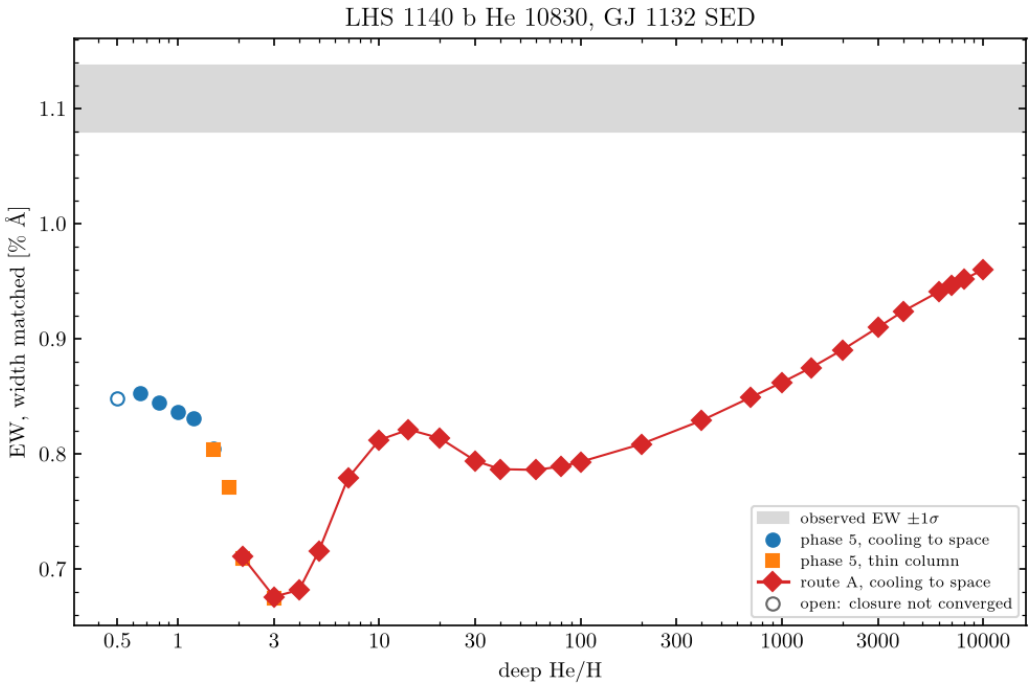


Figure 4: Width-matched He 10830 EW against deep He/H on the GJ 1132 spectrum with the full equation set (the table above), against the observed 1.1084 ± 0.0289 (grey). Figure of `route_heh_high/`, made by `compare_route_heh_high.py`.

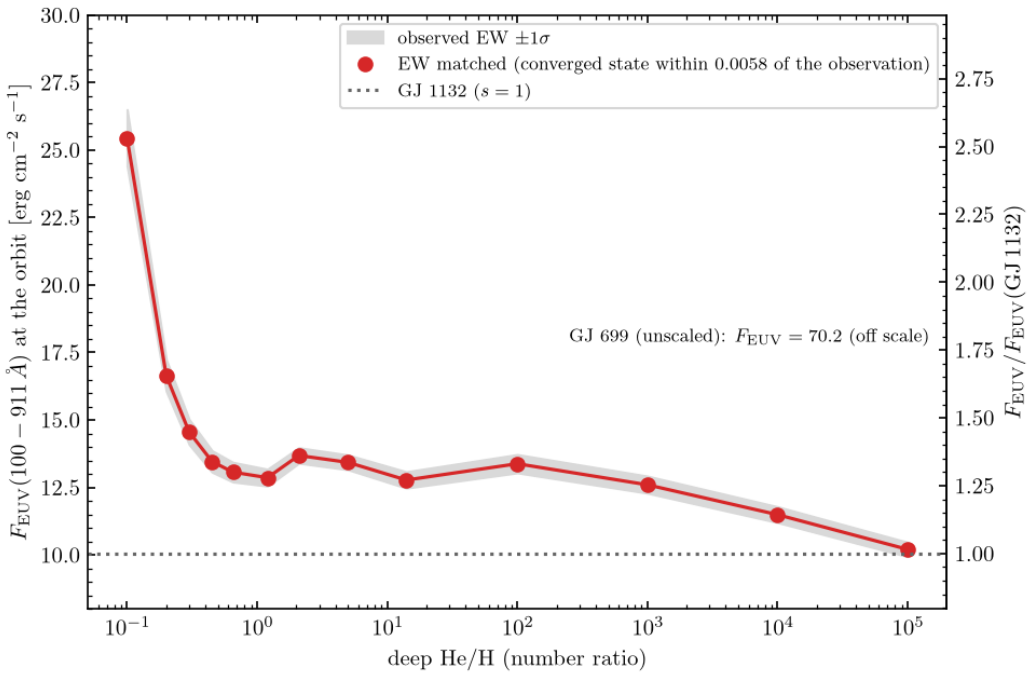


Figure 5: F_{EUV}(100–911 Å) at which the width-matched EW reaches the observed value, against deep He/H (the table above; grey, the observed 1σ). Right axis: the factor s over GJ 1132. Figure of `route_xuv_scaled/`, made by `compare_euv_match_vs_heh.py`.

Both clear the flux-spread criterion by four to eight decades and ONE row separates them. The two states differ by at most 1.9 per cent in any profile, and the difference sits in the composition and the temperature of the layer the He I 10830 line forms in (the metastable column's median radius is 6.4 R_p, its 5 to 95 per cent range 1.7 to 16.7 R_p), not in $\rho v r^2$, which agrees to 0.055 per cent everywhere. That is why the flux-spread criterion is blind to it.

It reaches a measured quantity. With the catalog's own selection the He I 10830 red-pair equivalent width moves from 1.374869 to 1.356745 % Å, a change of 1.32 per cent, and the red depth by 1.11 per cent, while $\log_{10} \dot{M}$ is 7.87 for both at the two decimals the code prints. For scale, the He/H rungs of this group stand 0.178 % Å apart in the equivalent width, so the shift is a tenth of the spacing the ladder measures a composition with.

Against the measurement it is smaller still. The observed red-pair equivalent width is 1.108 ± 0.029 % Å, integrated from the spectrum Cherubim et al. (2026) released, with the error of the same estimator under independent sample errors (our number, not the paper's; the tables and figures of this document were drawn with ± 0.030, the error before the estimator was given its own trapezoidal weights on 2026-09-29), and the two states differ by 0.018 % Å, which is 0.62 of one standard deviation. For a comparison with the data the two states are the same model: the refused one would be read as the same answer. What the certification buys is not a different observable but the statement that the state solves the elemental transport equation everywhere, which is what lets a later measurement, or a ladder that resolves 0.178 % Å in composition, rest on it.

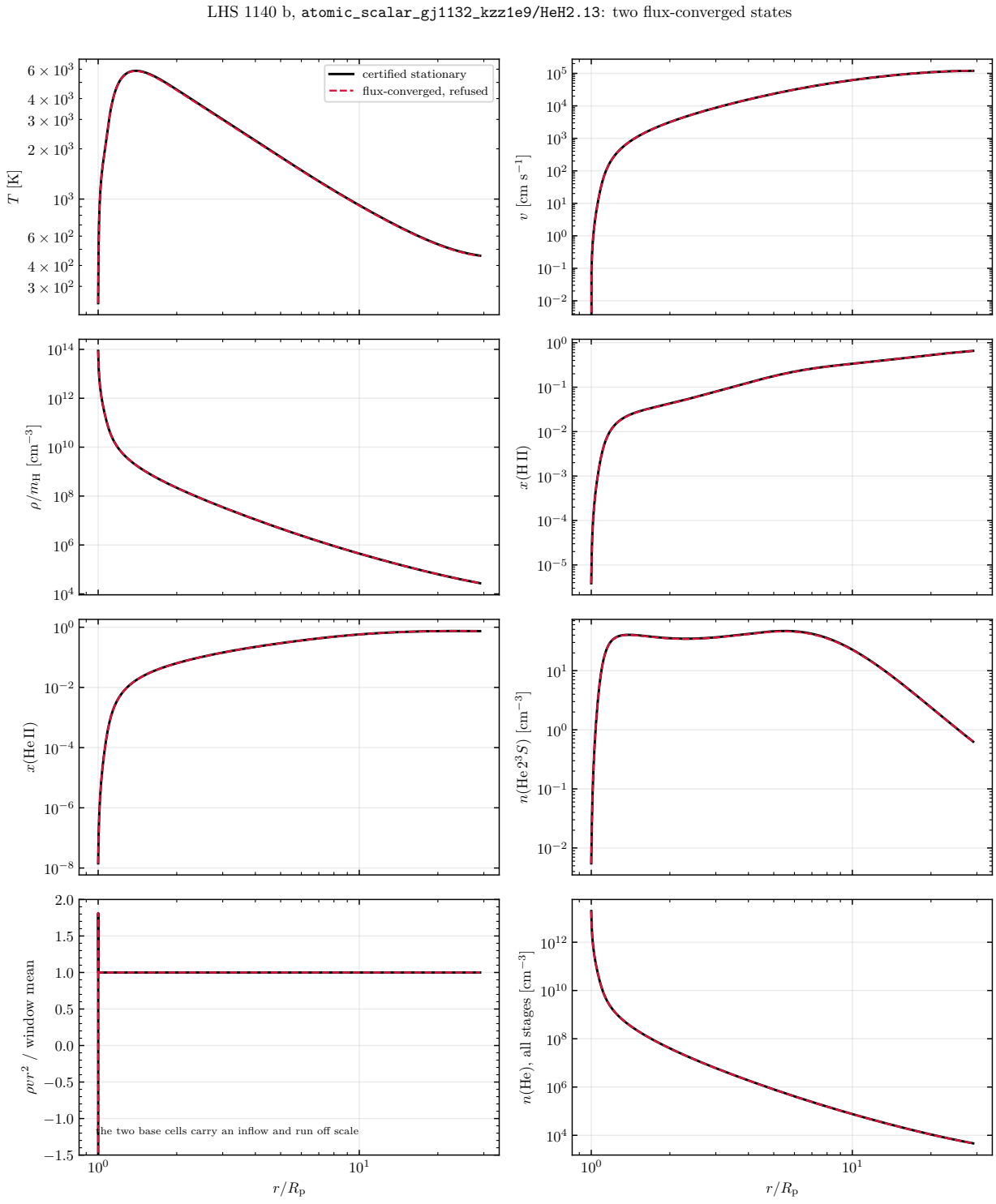


Figure 6: The certified stationary solution and the flux-converged refused state of the atomic `scalar_kzz1e9/HeH2.13`, as functions of radius. The last panel is the number density of all helium nuclei, He I + He II + He III, independent of ionization and level.

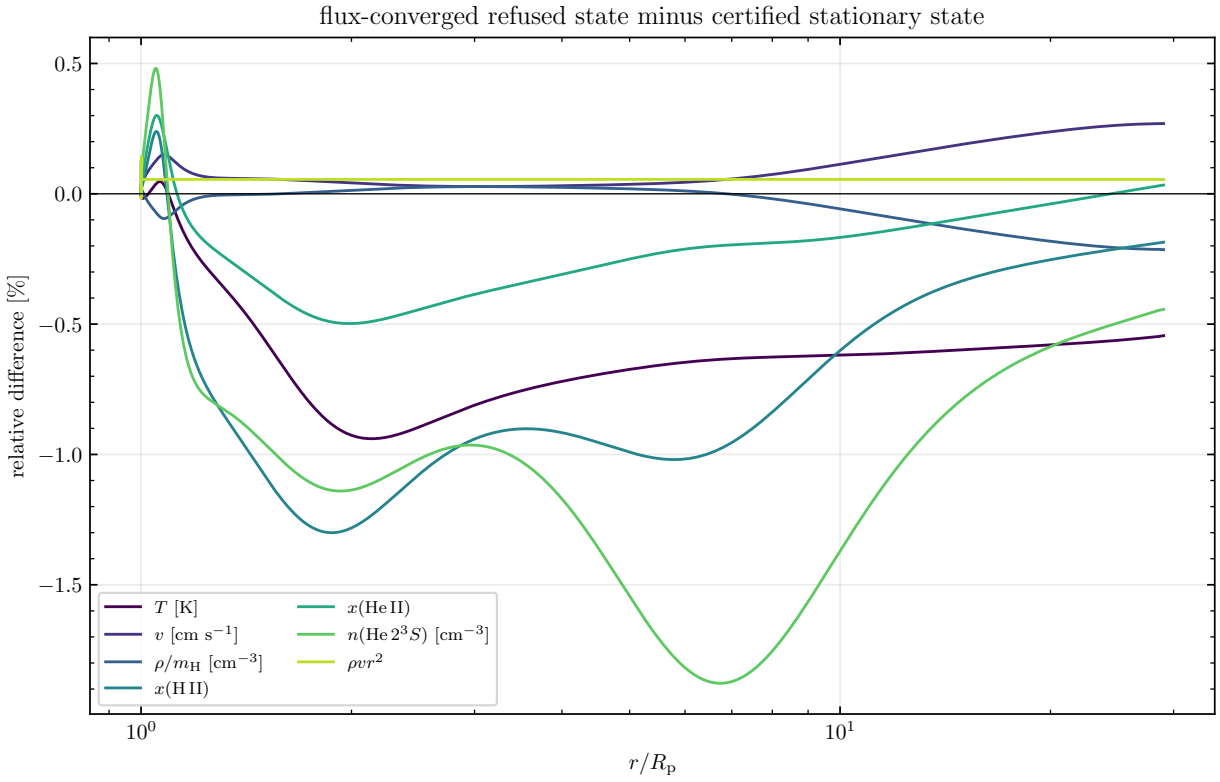


Figure 7: The same two states, as the relative difference of each quantity. The largest are the temperature at $r = 2.13\,R_p$ (–0.94 per cent), x(H II) at 1.87 (–1.30) and n(He 2^3S) at 6.74 (–1.88).

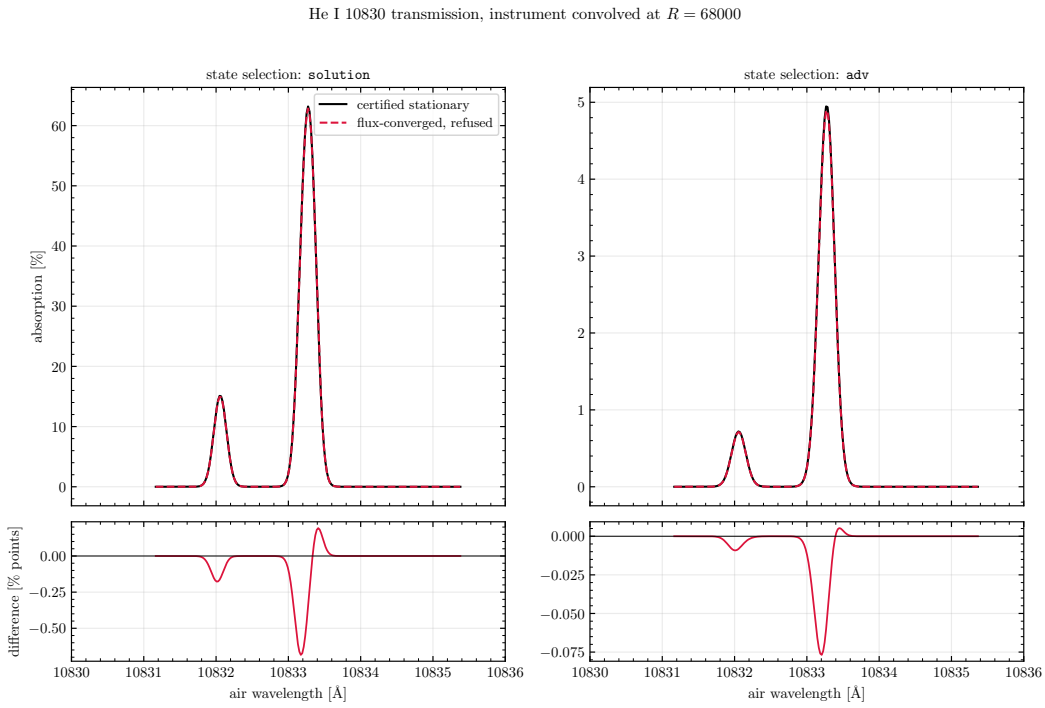


Figure 8: The He I 10830 transmission of the two states.